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Homework

A homework for class **MET CS 579** at Boston University, instructed by **Jae Lee**. Student Jin Yu Zhang submitted on **March 28, 2023**.

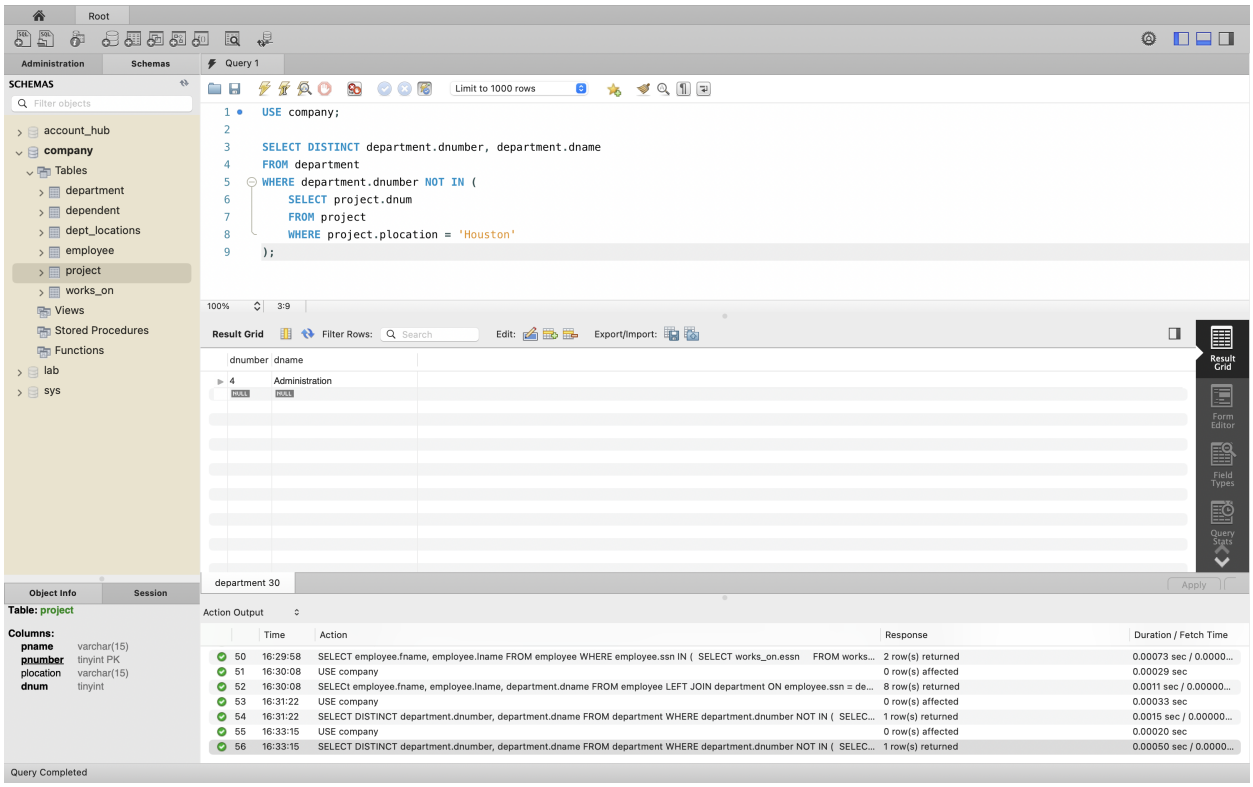
Problem 1

1

Show the department number and the department name of every department that controls a project that is not located in Houston:

```
USE company;

SELECT DISTINCT department.dnumber, department.dname
FROM department
WHERE department.dnumber NOT IN (
    SELECT project.dnum
    FROM project
    WHERE project.plocation = 'Houston'
);
```

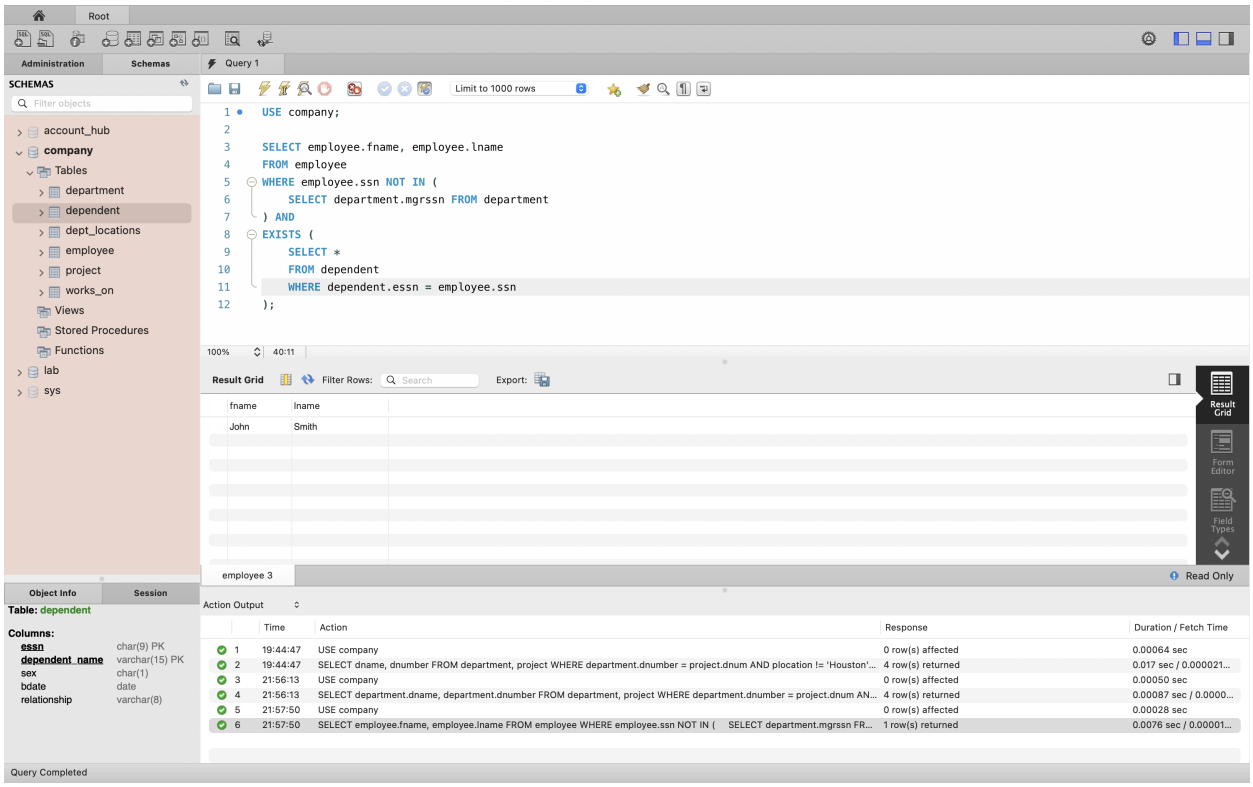


2

Show the first name and the last name of every employee who is a not manager and has a dependent:

```
USE company;

SELECT employee.fname, employee.lname
FROM employee
WHERE employee.ssn NOT IN (
    SELECT department.mgrssn FROM department
) AND
EXISTS (
    SELECT *
    FROM dependent
    WHERE dependent.essn = employee.ssn
);
```

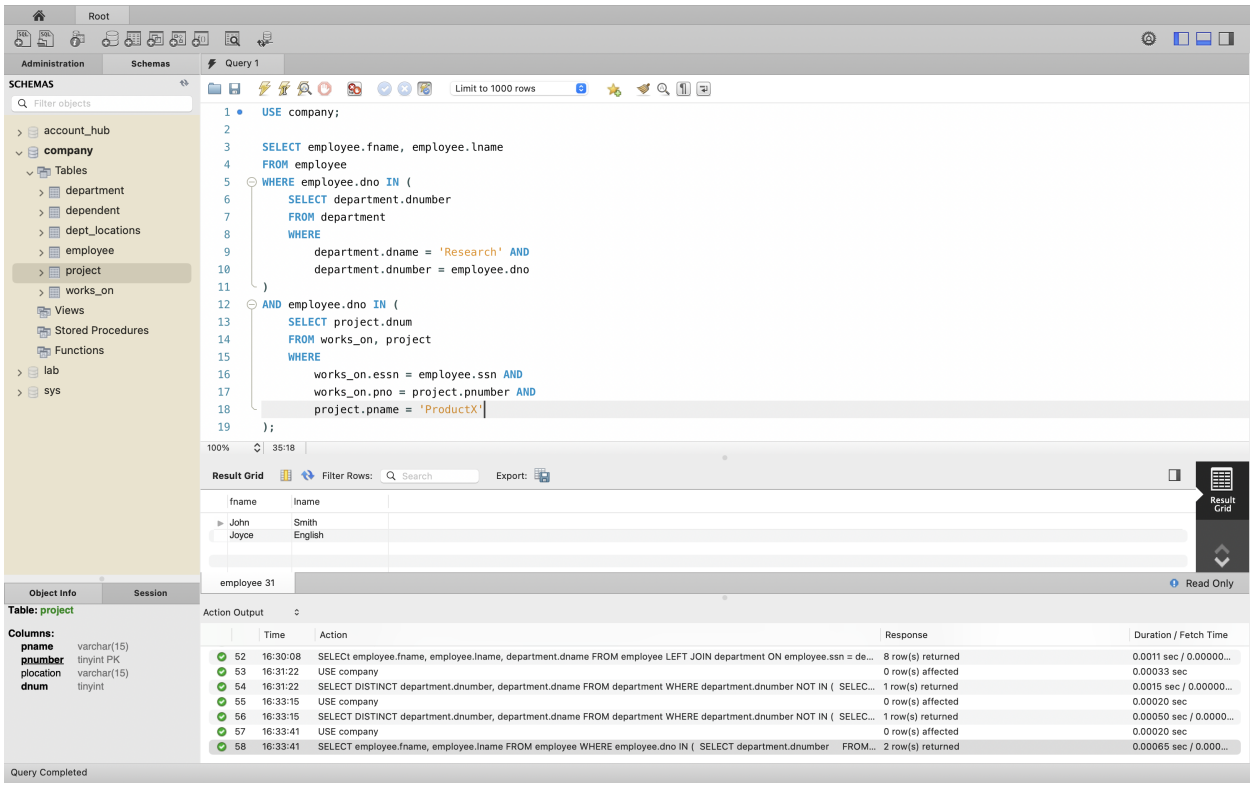


3

Show the first name and the last name of every employee who works for the Research department and works on the ProductX project:

```
USE company;

SELECT employee.fname, employee.lname
FROM employee
WHERE employee.dno IN (
    SELECT department.dnumber
    FROM department
    WHERE
        department.dname = 'Research' AND
        department.dnumber = employee.dno
)
AND employee.dno IN (
    SELECT project.dnum
    FROM works_on, project
    WHERE
        works_on.essn = employee.ssn AND
        works_on.pno = project.pnumber AND
        project.pname = 'ProductX'
);
```



4

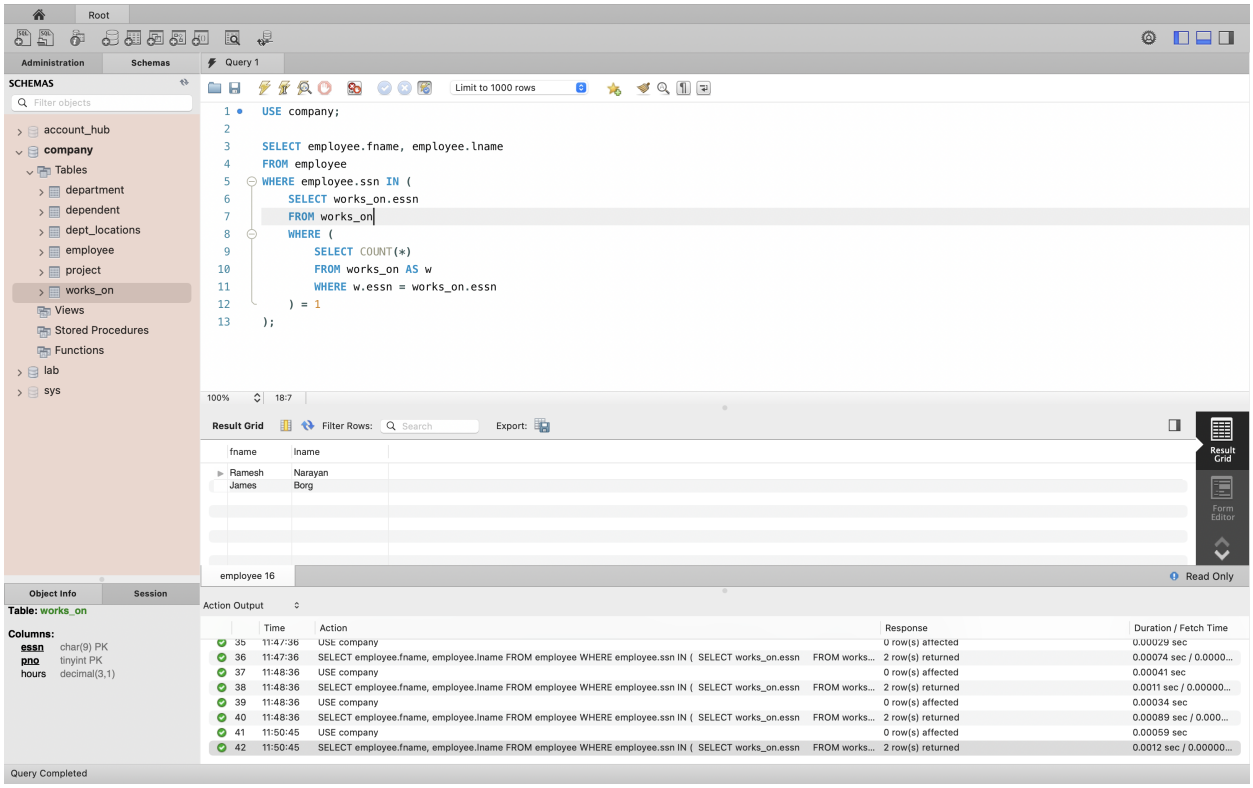
Show the first name and last name of every employee who works on only one project:

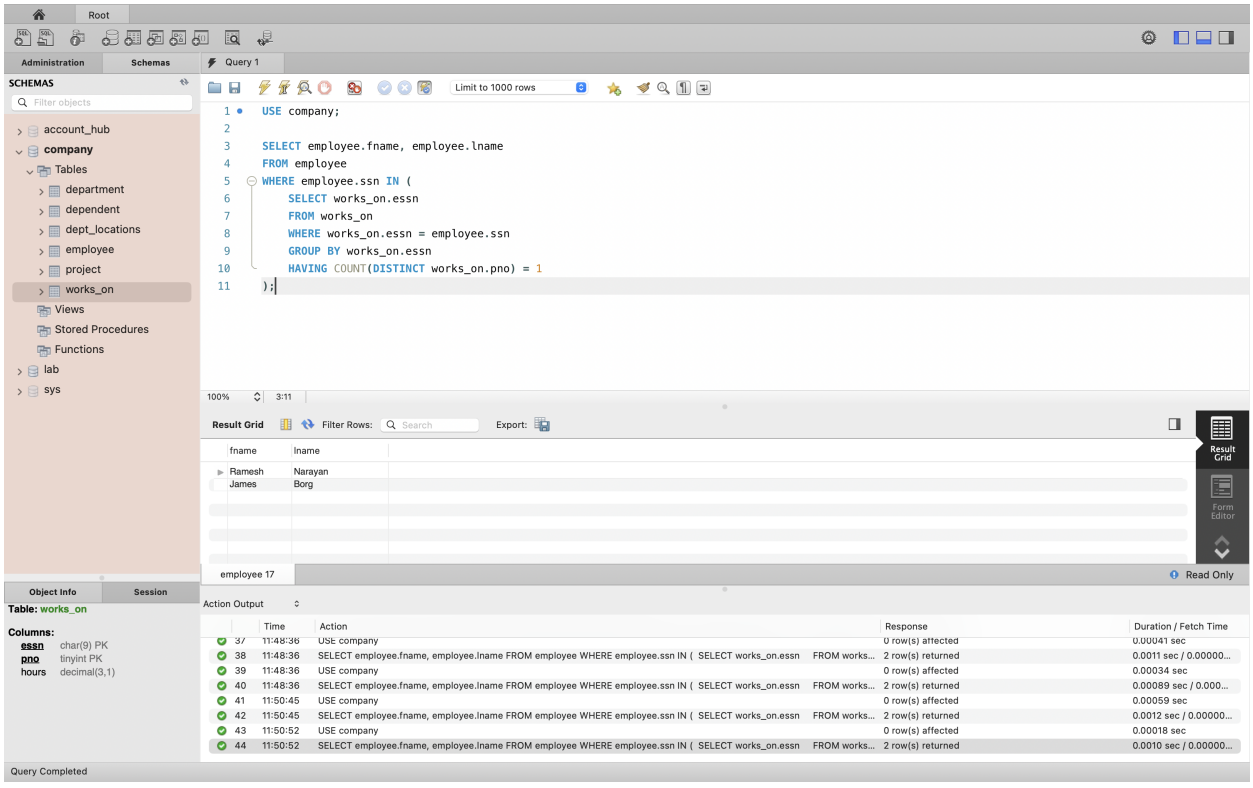
```
USE company;

SELECT employee.fname, employee.lname
FROM employee
WHERE employee.ssn IN (
    SELECT works_on.essn
    FROM works_on
    WHERE (
        SELECT COUNT(*)
        FROM works_on AS w
        WHERE w.essn = works_on.essn
    ) = 1
);
-- or
USE company;

SELECT employee.fname, employee.lname
FROM employee
WHERE employee.ssn IN (
    SELECT works_on.essn
    FROM works_on
    WHERE works_on.essn = employee.ssn
```

```
    GROUP BY works_on.essn
    HAVING COUNT(DISTINCT works_on.pno) = 1
);
```





5

Show the first name and the last name of every employee. If an employee is the manager of a department, then show the name of the department the employee manages too. The result must

be in a single relation:

```
USE company;

SELECt employee.fname, employee.lname, department.dname
FROM employee LEFT JOIN department ON employee.ssn = department.mgrssn;
```

